

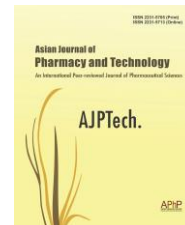
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REVIEW ARTICLE

Schedule T – Good Manufacturing Practice of Indian systems of medicine

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ABSTRACT:

On June 23, 2000, the Government of India issued GSR No. 560(E) under Rule 157 of the Drugs and Cosmetic Rules, and on March 7, 2003, GSR No. 198(E) revised Good Manufacturing Practices (GMP). These rules were made necessary for all new ASU drug manufacturing facilities on June 23, 2000, and for existing units on June 23, 2002. (2 years grace period was given for existing ASU units to obtain GMP certification). The criteria for production premises and hygienic conditions are outlined in Schedule T. The fundamental goal of implementing Schedule-T was to ensure that producers maintained a consistent level of hygiene. Good manufacturing practises must be followed at all times. For all the manufacturers of ASU drugs. Compliance to Good manufacturing Practices is mandatory for all the manufacturers of ASU drugs.

KEYWORDS: Good manufacturing practices for Ayurveda, siddha unani medicines.

INTRODUCTION:

The Good Manufacturing Practices (GMP) are prescribed as follows in Part I and Part II to ensure: -

1. Raw materials used in the manufacture of drugs are authentic, of prescribed quality and are free from contamination;
2. The manufacturing process is as has been prescribed to maintain the standards; Adequate quality control measures are adopted;
3. The manufactured drug which is released for sale is of acceptable quality;
4. To achieve the objectives listed above, each licensee shall evolve methodology and procedures for following the prescribed process of manufacture of drugs which should be documented as a manual and kept for reference and inspection

However, under IMCC Act 1970 registered Vaidyas, Siddhas and Hakeems who prepare medicines on their own to dispense to their patients and not selling such drugs in the market are exempted from the purview of Good Manufacturing Practices (GMP).

PART-I

1.1 Factory Premises:

1.2 The manufacturing plant should have adequate space for:

- (i). Receiving and storing raw material
- (ii). Manufacturing process areas
- (iii). Quality control section
- (iv). Finished goods store
- (v). Office
- (vi). Rejected goods/drugs store

1.1 General Requirements:

1.1 (A) Location and surrounding:

For manufacture of Ayurveda, Siddha and Unani medicines shall be so situated and shall have such construction as to avoid contamination from open sewerage, drain, public lavatory or any factory which produces disagreeable or obnoxious odour or fumes or

excessive soot, dust or smoke.

1.1 (B) Buildings:

The building used for factory shall be such as to permit production of drugs under hygienic conditions and should be free from cobwebs and insects/rodents. It should have adequate provision of light and ventilation. The floor and the walls should not be damp or moist. The premises used for manufacturing, processing, packaging and labelling will be in conformity with the provisions of the Factory Act. It shall be located so as to be:

- I. Compatible with other manufacturing operations that may be carried out in the same or adjacent premises.
- II. Adequately provided with working space to allow orderly and logical placement of equipment and materials to avoid the risk of mix-up between different drugs or components thereof and control the possibility of cross-contamination by other drugs or substances and Ensure that no manufacturing or quality control steps are missed. Designed, built, and maintained to keep insects and rodents from entering. Surfaces inside the building (walls, floors, and ceilings) must be smooth and free of cracks to allow for easy cleaning and disinfection. The walls of the room where the manufacturing processes are performed must be impenetrable to contamination and cleanable. The flooring must be smooth and even, with no gaps or crevices that allow dust or waste materials to collect.
- III. Appropriate drainage system in the processing area is provided. In the manufacturing area, the sanitary fittings and electrical fixtures must be proper and safe.
- IV. The furnace/Bhatti area could be covered with a tin roof and proper ventilation, but flies and dust should be avoided. There should be fire safety measures in place, as well as proper exits. Drying area- A separate area should be set aside for the drying of raw materials.

1.1 (C) Water Supply:

The water used in manufacture shall be pure and of potable quality. Adequate provision of water for washing the premises shall be made.

1.1 (D) Disposal of Waste:

From the manufacturing sections and laboratories the waste water and the residues which might be prejudicial to the workers or public health shall be disposed of after suitable treatment as per guidelines of pollution control authorities to render them harmless.

1.1 (E) Containers' Cleaning:

In factories where operations involving the use of containers such as glass bottles, vials and jars are conducted, there shall be adequate arrangement separated from the manufacturing operations for washing, cleaning and drying of such containers.

1.1 (F) Stores:

Storage should have proper ventilation and shall be free from dampness. It should provide independent adequate space for storage of different types of material, such as raw material, packaging material and finished products.

1. Raw Materials:

All raw materials procured for manufacturing will be stored in the raw materials store. The manufacture based on the experience and the characteristics of the particular raw material used in Ayurveda, Siddha and Unani system shall decide the use of appropriate containers which would protect the quality of the raw material as well as prevent it from damage due to dampness, microbiological contamination or rodent and insect infestation, etc. If certain raw materials require such controlled environmental conditions, the raw materials stores may be sub-divided with proper enclosures to provide such conditions by suitable cabinization. While designing such containers, cabins or areas in the raw materials store, care may be taken to handle the following different categories of raw materials:-

- (a) Raw material of metallic origin.
- (b) Raw material of mineral origin.
- (c) Raw material from animal source.
- (d) Fresh Herbs.
- (e) Dry Herbs or plant parts.
- (f) Excipients, etc.
- (g) Volatile oils/perfumes and flavours.
- (h) Plant concentrates /extracts and exudates/resins Each container used for raw material storage shall be properly identified with the label which indicates name of the raw material, source of supply and will also clearly state the status of raw material such as 'UNDER TEST' or 'APPROVED' or 'REJECTED'. The labels shall further indicate the identity of the particular supply in the form of Batch No. or Lot. No. and the date of receipt of consignment. All the raw materials shall be sampled and got tested either by the in-house Ayurvedic, Siddha and Unani experts (Quality control technical person) or by the laboratories approved by Government and shall be used only on approval after verifying. The rejected raw material should be removed from other raw materials store and should be kept in a separate room. Procedure of 'First in first out' should be adopted for raw materials wherever necessary. Records of the receipt, testing and approval or rejection and use of raw material shall be maintained.

2. Packaging Materials:

All packaging materials such as bottles, jars, capsules, etc. Shall be stored properly. All containers and closures shall be adequately cleaned and dried before packing the products.

3. Finished Goods Stores:

The finished goods transferred from the production area after proper packaging shall be stored in the finished goods stores within an area marked "Quarantine". After the quality control laboratory and the experts have checked the correctness of finished goods with reference to its packing/labelling as well as the finished product quality as prescribed, then it will be moved to 'Approved Finished Goods Stock' area. Only approved finished goods shall be dispatched as per marketing requirements. Distribution records shall be maintained as required. If any Ayurvedic, Siddha and Unani drug needs special storage conditions, finished goods store shall provide necessary environmental requirements.

1.1 (G) Working Space:

The manufacturing area shall provide adequate space (manufacture and quality control) for orderly placement of equipment and material used in any of the operations for which these are employed so as to facilitate easy and safe working and to minimize or to eliminate any risk of mix-up between different drugs, raw materials and to prevent the possibility of cross-contamination of one drug by another drug that is manufactured, stored or handled in the same premises.

1.1 (H) Health, Clothing, Sanitation and Hygiene of Workers:

All workers employed in the Factory shall be free from contagious diseases. The clothing of the workers shall consist of proper uniform suitable to the nature of work and the climate and shall be clean. The uniform shall also include cloth or synthetic covering for hands, feet and head wherever required. Adequate facilities for personal cleanliness such as clean towels, soap and scrubbing brushes shall be provided. Separate provision shall be made for lavatories to be used by men and women, and such lavatories shall be located at places separated from the processing rooms. Workers will also be provided facilities for changing their clothes and to keep their personal belongings.

1.1(I) Medical Services- The manufacturer shall also provide:

- a) Adequate facilities for first aid
- b) Medical examination of workers at the time of employment and periodical check-up thereafter by a physician once a year, with particular attention being devoted to freedom from infections. Records thereof shall be maintained.

1.1(J) Machinery and Equipment:

For carrying out manufacturing depending on the size of operation and the nature of product manufactured, suitable equipment either manually operated or operated semi-automatically (electrical or team based) or fully automatic machinery shall be made available. These may include machines for use in the process of manufacture such as crushing, grinding, powdering, boiling, mashing, burning, roasting, filtering, drying, filling, labelling and packing, etc. To ensure ease in movement of workers and orderliness in operations a suitably adequate space will be ensured between two machines or rows of machines. These machinery and equipment and machinery recommended is indicated in Part II-A. Proper standard operational procedures (SOPs) for cleaning maintaining and performance of every machine should be laid down.

1.1(K) Batch Manufacturing Records:

The licensee shall maintain batch manufacturing record of each batch of Ayurvedic, Siddha and Unani drugs manufactured irrespective of the type of product manufactured (classical preparation or patent and proprietary medicines). Manufacturing records are required to provide and account of the list of raw materials and their quantities obtained from the store, tests conducted during the various stages of manufacture like taste, colour, physical characteristics and chemical tests as may be necessary or indicated in the approved books of Ayurveda, Siddha and Unani mentioned in the First Schedule of the Drugs and Cosmetics Act, 1940 (23 of 1940). These tests may include any in-house or pharmacopoeial test adopted by the manufacturer in the raw material or in the process material and in the finished product. These records shall be duly signed by Production and Quality Control Personnel respectively. Details of transfer of manufactured drug to the finished products store including dates and quantity of drugs transferred along with record of testing of the finished product, if any, and packaging, records shall be maintained. Only after the manufactured drugs have been verified and accepted quality shall be allowed to be cleared for sale. It should be essential to maintain the record of date, manpower, machine and equipments used and to keep in process record of various shodhana, bhavana, buring in fire and specific grindings in terms of internal use.

1.1(L) Distribution Records:

Records of sale and distribution of each batch of Ayurveda, Siddha and Unani Drugs shall be maintained in order to facilitate prompt and complete recall of the batch, if necessary. The duration of record keeping should be the date of expiry of the batch, Certain categories of Ayurvedic, Siddha and Unani medicines like Bhasma, Rasa, Kupi-pakva, Parpati, Sindura,

Karpu/Uppu/Puram, Kushta, Asava-arista, etc. do not have expiry date, in contrast their efficacy increases with the passage of time. Hence, records need to be maintained up to 5 years of the exhausting of stock.

1.1(M) Record of Market Complaints:

Manufacturers shall maintain a register to record all reports of market complaints received regarding the products sold in the market. The manufacturer shall enter all data received on such market complaints, investigations carried out by the manufacturers regarding the complaint as well as any corrective action initiated to prevent recurrence of such market complaints shall also be recorded. Once in a period of six months the manufacturer shall submit the record such complaints to the Licensing Authority. The Register shall also be available for inspection during any inspection of the premises. Reports of any adverse reaction resulting from the use of Ayurvedic, Siddha and

Unani drugs shall also be maintained in a separate register by each manufacturer. The manufacturer shall investigate any of the adverse reaction to find if the same is due to any defect in the product, and whether such reactions are already reported in the literature or it is a new observation.

1.1(N) Quality Control:

Every licensee is required to provide facility for quality control section in his own premises or through Government-approved testing laboratory. The test shall be as per the Ayurveda, Siddha and Unani pharmacopoeial standard. Where the tests are not available, the test should be performed according to the manufacturer's specification or other information available. The quality control section shall verify all the raw materials, monitor in process, quality checks and control the quality of finished product being released to finished goods store/warehouse. Preferably for such quality control there will be a separate expert. The quality control section shall have the following facilities:

- (1) There should be 150 sq feet area for quality control section.
- (2) For identification of raw drugs, reference books and reference samples should be maintained.
- (3) Manufacturing record should be maintained for the various processes.
- (4) To verify the finished products, controlled samples of finished products of each batch will be kept till the expiry date of product.
- (5) To supervise and monitor adequacy of conditions under which raw materials, Semi-finished products and finished products are stored.
- (6) Keep record in establishing shelf life and storage requirements for the drugs.
- (7) Manufacturers who are manufacturing patent

proprietary Ayurveda, Siddha and Unani medicines shall provide their own specification and control references in respect of such formulated drugs.

- (8) The record of specific method and procedure of preparation, that is, "Bhavana", "Mardana" and "Putta" and the record of every process carried Out by the manufacturer shall be maintained.
- (9) The standards for identity, purity and strength as given in respective pharmacopoeias of Ayurveda, Siddha and Unani systems of medicines published by Government of India Shall be complied with.
- (10) All raw materials will be monitored for fungal, bacterial contamination with a View to minimize such contamination.
- (11) Quality control section will have a minimum of
 - a) One person with Ayurveda/Unani/Siddha qualification recognized under Schedule II of Indian Medicine Central Council Act, 1970. Two other Persons, one each with Bachelor qualification in Botany/Chemistry/Pharmacy could be on part-time or on contractual basis.
 - b) The manufacturing unit shall have a quality control section as explained under Section 35(ii). Alternatively, these quality control provisions will be met by getting testing, etc., from a recognized laboratory for Ayurveda, Siddha and Unani drugs; under Rule 160-A of the Drugs and Cosmetics Act. The manufacturing company will maintain all the record of various tests got done from outside recognized laboratory.
 - c) List of equipment recommended is indicated in Part II-C.

1.2 Requirement for Sterile Product:

(A) Manufacturing Areas:

For the manufacture of sterile Ayurvedic, Unani and Siddha drugs, separate enclosed areas specifically designed for the purpose shall be provided. These areas shall be provided with air locks for entry and shall be essentially dust free and ventilated with an air supply. For all areas where aseptic manufacture has to be carried out, air supply shall be filtered through bacteria retaining filters (HEPA Filters) and shall be at a pressure higher than in the adjacent areas. The filters shall be checked for performance on installation and periodically thereafter the record of checks shall be maintained. All the surfaces in sterile manufacturing areas shall be designed to facilitate cleaning and disinfection. For sterile manufacturing routine microbial counts of all Ayurvedic, Siddha and Unani drug manufacturing areas shall be carried out during operations. Results of such count shall be checked against established in-house standards and record maintained. Access to manufacturing areas shall be restricted to minimum number of authorized personnel. Special procedure to be

followed for entering and leaving the manufacturing areas shall be written down and displayed. For the manufacturing of Ayurvedic, Siddha and Unani drug that can be sterilized in their final containers, the design of the areas shall preclude the possibility of the products intended for sterilization being mixed with or taken to be products already sterilized. In case of terminally sterilized products, the design of the areas shall preclude the possibility of mix-up between non-sterile products.

(B) Precautions against contamination and mix:

- Carrying out manufacturing operations in a separate block of adequately isolated building or operating in an isolated enclosure within the building,
- Using appropriate pressure differential in the process area.
- Providing a suitable exhaust system.
- Designing laminar flow sterile air system for sterile products.
- The germicidal efficiency of UV lamps shall be checked and recorded Indicating the burning hours or checked using intensity.
- Individual containers of liquids and ophthalmic solutions shall be examined against black-white

background fitted with diffused light after filling to ensure freedom from contamination with foreign suspended matter.

- Expert technical staff approved by the Licensing Authority shall check and compare actual yield against theoretical yield before final distribution of the batch. All process controls as required under master formula including room temperature, relative humidity, volume filled, leakage and clarity shall be checked and recorded.

Part-II:

A. List of Recommended Machinery, Equipment and Minimum Manufacturing Premises Required for the Manufacture of Various Categories of Ayurvedic, Siddha System of Medicines:

One machine indicated for one category of medicine could be used for the manufacturing of Other category of medicine also. Similarly some of the manufacturing areas like powdering,

Furnace, packing of liquids and Avaleha, Paks, could also be shared for these items.

Table.1 list of recommended machinery, equipment and minimum manufacturing premises required for the manufacture of various categories of Ayurvedic, siddha system of medicines

Sl. No	Category of Medicine	Minimum manufacturing space required	Machinery/equipment recommended
		1200 Square feet covered area with separate cabins or partitions for each Activity. If Unani medicines are manufactured in same premises an additional area of 400 sq. feet will be Required.	
1.	Anjana/Pisti	100 sq. feet.	Karel/mechanized/motorized, karel.
2.	Churna / Nasya/ Manjan/Lepa/ Kwath Churn	200 sq feet	Grinder/disintegrator/Pulveriser/ Powder mixer/sieves/shifter.
3.	Pills/Vati /Gutika Matirai and tablets	100 sq. feet	Ball Mill, Mass mixer/powder mixer, Granulator, drier, tablet compressing machine, pill/vati cutting machine, stainless steel trays/container for storage and sugar coating, polishing pan in case of sugar-coated tablets, mechanised chattoo (for mixing guggulu) where required.
4.	Kupi pakava/Ksara/ Parpati/LavanaBhasma Satva/Sindura Karpu/ Uppu / Param	150 sq. feet.	feet Bhatti, Karahi/Stainless steel Vessels/Patila Flask, Multani Matti/Plaster of Paris, Copper Rod, Earthen container, Gaj Put Bhatti, Muffle furnace (Electrically operated) End/Edge Runner, Exhaust Fan, Wooden/S.S.Spatula.
5.	Kajal	100 sq. feet	Earthen lamps for collection of Kajal, Triple Roller Mill, End Runner, Sieves, S.S. Patila, Filling/ packing and manufacturing room should be provided with exhaust fan and ultra violet lamps.
6.	Capsules	100 sq.	Air Conditioner, De-humidifier, hygrometer, thermometer, Capsule filling machine and chemical balance.
7.	Ointment/Marham Pasai	100sq.	Tube filling machine, Crimping Machine/Ointment Mixer, End Runner/ Mill (Where required) S.S. Storage Container S.S.Patila.
8.	Pak/Avaleh/Khand/ Modak/Lakayam	100 sq. feet	Feet Bhatti section fitted with exhaust fan and should be fly proof, Iron Kadahi/S.S. Patila and S.S. Storage container.
9.	Panak, Syrup / Pravahi Kwath Manapaku.	150 sq. feet	Tincture press, exhaust fan fitted and fly proof, Bhatti section, Bottle washing machine, filter press / Gravity filter, liquid filling machine, P.P. Capping Machine.
10.	Asava / Arishta	200 sq.ft	Same as mentioned above. Fermentation tanks, containers and

			distillation plant where necessary, Filter Press.
11.	Sura	100 sq. ft	Same as mentioned above plus Distillation plant and Transfer pump.
12.	Ark Tinir	100 sq. ft	Maceration tank, Distillation plant, Liquid filling tank with tap / Gravity filter/Filter press, Visual inspection box.
13.	Tail/Ghrit Ney	100 sq. ft	Bhatti, Kadahi/S.S. Patila S.S.Storage Containers, Filtration equipment, filling tank with tap/Liquid filling machine.
14.	Aschyotan / Netra Malham Panir/Karn Bindu/Nasa-bindu	100 sq. ft	Hot air oven electrically heated with thermostatic control, kettle gas or electrically heated with suitable mixing arrangements, collation mill, or ointment mill, tube filling, equipment, mixing and storage tanks of stainless steel or of other suitable material sintered glass funnel, seitz filter or filter candle, liquid filling Equipment, autoclave.
15.	Each manufacturing unit will have a separate area for Bhatti, furnace boilers, puta, etc. This will have proper ventilation, Removal of smoke, prevention of flies, insets, dust etc. The furnace section could have tin roof.	200 . ft	

Table2. List of machinery, equipment and minimum manufacturing premises required for the manufacture of various categories of unani system of medicines

Sl. No	Category of Medicine	Minimum manufacturing space required	Machinery/equipment recommended
(1)	(2)	(3)	(4)
1.		1200 square feet covered area with separate cabins, partitions for each Activity. If Ayurveda / Siddha, Medicines are also manufactured in same premises an addition area of 400 square feet will be required	
2.	Itrifal Tiryo / majoon / Laooq / Jawarish Khamiras	100 Sq. feet	Grinder/ Pulveriser, Sieves, powder mixer (if required), Stainless Steel Patilas, Bhatti and other accessories, Planter mixer for Khamiras
3.	Araq.	100 Sq. feet	Distillation Plant (garembic) Stainless Steel storage tank, boiling vessel, gravity filter, bottle filling machine, bottle washing machine, bottle drier
4.	Habb (Pills) and tablets	100 Sq. feet	Ball Mill, Mass Mixer/Powder mixer, Granulator drier, tablet compressing machine, pill/vati cutting machine, stainless steel trays/ container for storage and in the case of sugar coated tablets, mechanized chattoo, (for mixing guggulu) where required.
5.	Sufoof (Powder)	100 Sq. feet	Grinder / Pulveriser, Sieves, Trays, Scoops, Powder mixer (where required).
6.	Raughan (oils) (Crushing and boiling)	100 Sq. feet	Oil expeller, Stainless Steel Patilas oil filter bottle, filling machine, bottle drier, bhatti.
7.	Shiyaf, Surma, Kajal	100 Sq. feet	End runner, mixing Stainless Steel Vessel
8.	Marham, Zimad Ointment)	100 Sq. feet	Karal, Bhatti, End runner, Grinder, Pulveriser, Triple Roller Mill (if needed).
9.	Qurs (Tab)	100 Sq. feet	Grinder / Pulveriser, Sieves, Powder mixer (where needed), Granulator, Drier, Tablet Compressing Machine, Die punches Trays, Disintegration apparatus, Balance with weights, Scoops, Sugar Coating Pan, Polishing pan, Heater.
10.	Kushta	100 Sq. feet	Bhatti, Kharal, Sil Batta, Earthen pots.
11.	Murabba	100 Sq. feet.	Aluminium Vessels 50-100 kgs. Capacity, Gendna, Bhatti.
12.	Capsule	100 Sq. feet	ulveriser, Powder mixer (where needed), capsule filling machine, Air conditioner, Dehumidifier Balance with weights, storage containers, glass
13.	Sharbat and Jushanda	100 Sq. feet	Tinctum Press, exhaust fan fitted, Bhatti section, Bottle washing machine, Filter Press, Gravity filter, Liquid filling tank with tap/liquid filling machine, PP capping

			machine, air over electrically heated with thermostatic control, kettle
14.	Qutoor Chasm and Marham (Eye drops eye ointment)	100 Sq. feet	Hot air oven electrically heated with thermostatic control, kettle.
15.	Each Manufacturing unit will have a separate area for Bhatti, fumances, boiler, putta, etc. This will have proper ventilation, removal of smoke, prevention of files, insects, dust, etc.	200 . feet	

B. List of Machinery, Equipment and Minimum Manufacturing Premises Required For The Manufacture Of Various Categories Of Unani System Of Medicines:

One machine indicated for one category of medicine could be used for the manufacturing of other category of machine also. Similarly some of the manufacturing areas like powdering, furnace, packing of liquids could also be shared for these items.

C. List of Equipment Recommended for in-House Quality Control Section:

(Alternatively, unit can get testing done from the Government approved laboratory)

(I) Chemistry section:

1. Alcohol Determination Apparatus (complete set)
2. Volatile Oil Determination Apparatus
3. Boiling Point Determination Apparatus
4. Melting Point Determination Apparatus
5. Refractometer
6. Polarimeter
7. Viscometer
8. Tablet Disintegration Apparatus
9. Moisture Meter
10. Muffle Furnace
11. Electronic Balance
12. Magnetic Stirrer
13. Hot Air Oven
14. Refrigerator
15. Glass/Steel Distillation Apparatus
16. LPG Gas Cylinders with Burners
17. Water Bath (Temperature controlled)
18. Heating Mantles/ Hot Plates
19. TLC Apparatus with all accessories (Manual)
20. Paper Chromatography apparatus with accessories
21. Sieve size 10 to 120 with Sieve shaker
22. Centrifuge Machine
23. Dehumidifier
24. pH Meter
25. Limit Test Apparatus

(II) Pharmacognosy Section:

1. Microscope Binocular
2. Dissecting Microscope
3. Microtome
4. Physical Balance

5. Aluminium Slide Trays
6. Stage Micrometer
7. Camera Lucida (Prism and Mirror Type)
8. Chemicals, Glassware etc.

D. Supplement Guidelines for Manufacturing of Rasauhadhies or Rasamarunthukal and Kushtajat (Herbomineral-Metallic Compounds) of Ayurveda, Siddha and Unani Medicines:

These guidelines are intended to complement those provided above and should be read in conjunction with the parent guidelines. The supplementary guidelines are to provide general and minimum technical requirements for quality assurance and control in manufacturing Rasaushadhis or Rasamarunthukal and Kushtajat (Herbo-mineral-metallic formulations). These supplementary guidelines deal with Bhasmas, Sindura, Pishti, Kajjali, Khalviya Ras, Kupipakwa, Rasayan, Parpati, Potali Rasa, Satwa (of Metals and Minerals origin) Druti Parpam, Karpu, and Kushta etc. used in Ayurvedic, Siddha and Unani Systems of medicine. The supplementary GMP guidelines for Rasaushadhi or Rasamarunthukal and Kushtajat are needed to establish the authenticity of raw drug, minerals and metals, in process validation and quality control parameters to ensure that these formulations are processed and prepared in accordance with classical texts and for which safety measures are complied. Only those manufacturing units which have Good Manufacturing Practices for ASU drugs and supplementary certificate for Rasaushadhi or Rasamarunthukal and Kushtajat formulations shall be allowed to manufacture the same. Supplementary Good Manufacturing practices Certificate for Rasaushadhis shall be issued by the State Licensing Authority only after thorough inspection by an expert team including Rasashastra experts nominated by the Department of AYUSH.

2. Manufacturing Process Areas:

For the Manufacturing of Bhasma and Kupipakawa and Rasaaushadhi preparations made from metals and minerals the following specific areas shall be provided, which should be completely segregated from the production areas used for preparations of plants and animals by product based formulations to avoid cross contamination. The following exclusive areas are required for Rasaushadhis or Rasamaraunthukul and

Kushtajat.

(a) Bhatti or Heating Devise Section for Bhasma and Rasaaushadhi:

100 sq. feet for heating, burning, putta and any heat related work with proper ventilation, exhaust and chimney. This could be tin shed also.

(b) (Grinding, Drying and Processing Section for Bhasma and Rasaaushadhi:

10sq. feet (Manual or mechanical, oven etc.). Drying may be done in a space which is covered by glass or other transparent material to allow entry of sunrays on the material to keep for the purpose. If drying is being in oven the temperature of the same may be selected specific temperature.

(c) Rasaushadhis Related Store:

100 sq. ft. The size and Dimensions of each Bhatti section would be so designed it suit the batch size or quantity of materials to be processed, keeping in mind the processed, keeping in mind the processing is done as per the conditions of Drugs and Cosmetics Act mentioned under Schedule. In additions to the fuel prescribed in the schedule books namely coal, fire woods, cow dung cakes etc., use of other heating devices e.g. electrical heating, oil or gas fired furnace and other may be employed so as to provide the required temperature as per the nature of material and object of heating. Depending on the formulation being manufactured, manufacturers may adopt aerobic or anaerobic process. Properly baked and clean earthen pots of other crucibles and glass containers or appropriate design shall be used. The manufacturing areas should be designed with special attention to process the products that generate toxic fumes like SO₂, arsenic and mercury vapour, etc. When heating and boiling is necessary, suitable ventilation and air exhaust flow mechanism should be provided to prevent accumulation of unintended fumes and vapours. Such areas may be provided with properly designed chimneys or ducts fitted with exhaust systems and suitable scrubbing system to remove fumes and smokes, so that safety of personnel and environment is taken care of. Since processing of Rasaaushadhi may introduce heavy metal contamination and cross contamination etc., therefore, cleaning of equipment is particularly important after every process by using appropriate cleaning agent which should not react with material of equipment and must be free unwanted properties e.g. corrosiveness.

Records shall be maintained specially for temperatures attained during the entire process of Bhasmikaran, while employing different kinds of classical puta, furnace using oil, gas or electricity. Appropriate temperature measuring instrument should be employed such as pyrometer and, pyrograph for manual reading or

recording by heat sensors, connected to computer, as the case may be.

In order to handle large quantities, appropriate technology like use of hand operated extruders for making chakrikas or pellets may be adopted. However, such equipment made of aluminium or its alloys should not be used. Access to manufacturing areas shall be restricted to minimum number of authorised personal only.

Quality Control:

A. In Process Quality Control:

The registers as indicated below should exclusively be maintained for ready reference:

(a) Shodhan Register with following details:

- (1) Sl. No.
- (2) Batch no. and Size
- (3) Date, time and duration
- (4) Name of the Raw-material with Quality reference and quantity
- (5) Quantity of Shodhana Dravya
- (6) Book reference followed
- (7) Methodology

(b) Bhavana and Putta Register with following details:

- (1) Sl. No.
- (2) Batch no. and Size
- (3) Date, time and duration
- (4) Name of the material with Quantity of starting materials
- (5) Quantity of nirvapy Dravya
- (6) Quantity of Bhavana Dravya
- (7) Date and Time of Starting and completion of Bhavana or Mardana and duration
- (8) Type and Number of Puttas
- (9) Time and Date of completion Puttas
- (10) Colour and texture of the products or standards
- (11) In process tests followed (Bhasma Pariksha and any other tests)
- (12) In case heating at a particular temperature is required, record of attainment of that temperature.

(c) Grinding Records Register: (Finished Products/Intermediate Procedure):

- (1) Sl. No.
- (2) Batch no. and Size
- (3) Date, time and duration
- (4) Name of the material with Quantity
- (5) Name of the equipment (SS/granite)
- (6) Duration of grinding
- (7) Repeat the grinding if required (number of repetition)

(d) Packing details:

- (1) Name of Rasaushadhi
- (2) Type of Dosage Form (e.g. Powder, pill tablet etc.)
- (3) Weight of Rasaushadhis in each unit.

B. Product Quality Control:

The specifications for finished Rasaushadhi are primarily intended to define the quality rather than to establish full characterization, and should focus on those characteristics found to be useful in ensuring the quality. Consistent quality for Rasaushadhi can only be assured if the starting material-metals and minerals are used of pharmacopoeial standards. In some cases more detailed information may be needed on aspects of their process. The manufacture will ensure in-house standards for the uniform quality of products.

Quality testing will be carried out as per official Pharmacopoeia or Schedule books for tests namely, colour, taste, varitaratwa, Rekhapurnatwa, Laghutva, Nirudhumatwa, Dntagree kachakacha, Niruttha, Apunarbhava and Nischandratwa.

The Particle size of products should be tested adopting microscopic fitted with micrometre of particle size analyser or any appropriate other techniques. Required physio-chemical characterization of the product should be undertaken by appropriate analytical equipment. The Standard Manufacturing Process of the product should be evolved/follow up. The disintegration time of pills-vati and tablets should also be recorded.

4. Product recalls:

Literature inserted inside the products package should indicate the name, address of the manufacturing unit or email or telephone number for reporting of any adverse drug reaction by physicians or patients. On receipt of such Adverse Drug Reaction report, it will be the responsibility of the manufacturer to ensure the recall the product from the market. Standard operating procedures (SOP) should be included for storage of recalled Rasaushadhis in a secure segregated area, complying with the requirements specified for storage, till their final disposal.

5. Medical Examination of the employees:

Employees engaged in manufacturing should be medically examined periodically at least once a year for any adverse effect of the drug during manufacturing process for which necessary investigations may be carried out for ensuring that there is no effect of material on the vital organs of the employees. Annual examination reports of the employees shall be made available to statutory inspectors during Good Manufacturing Practices inspections.

6. Self- Inspection:

The release of Rasaushadhis should be under the control of a person who has been trained in the specific features of the processing and quality assurance of Rasaushadhis Personnel dealing with the production and quality assurance of Rasaushadhis manufacturing section should have an adequate training in the specific subject of Rasaushadhis manufacturing.

7. Dosage form of Rasaushadhis:

The Rasaushadhis may be made into an acceptable dosage forms such as, churna, vati, guti, tablet, capsule etc. after adding suitable permissible fillers or binding agents as permissible under the Ayurvedic Pharmacopoeia of India or Indian Pharmacopoeia as updated from time to time. In such cases the label must indicate the Ayurveda/Siddha/ Unani medicine in one Tablet or Pill or Capsule in addition to the filters. The crystalline product may be grinded before packing in the individual dispensing size. All the Rasaushadhis or Rasamaraunthukul and Kushtajat shall be packed in a dosage form which is ready for use for the consumer. Grinding and weighting of individual dose of potentially poisonous products will not be permissible in patient consumer pack. This arrangement may reduce the Adverse Drug Reaction of Rasaushadhis which takes place due to dose variation. However for hospital bulk pack, it will not be applicable and label will not be applicable and label will clearly indicate the "Hospital Pack".

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